Zhen Liu

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Education

University of California, Berkeley Berkeley, CA

Bachelor of Science in Electrical Engineering & Computer Sciences *GPA: 3.56*

Coursework: Data Structures (Java), Discrete Math & Probability Theory, Circuits, Signals & Systems, Quant Finance, Computer Architecture (C), Random Processes, Optimization Models

Experience

Software Engineer Intern Aug 2025 – Present

AfterQuery (YC W25) *Remote*

- Engineered the frontend of a large-scale LLM evaluation platform for financial analysis, enabling real-time response comparison and user voting using Next.js, Tailwind CSS, and TypeScript
- Developed LLM evaluation pipelines to benchmark 20 large language models on accuracy and reasoning for financial analysis tasks, improving model selection and deployment decisions
- Re-architected the Elo rating system using Firebase and Google Cloud, reducing database reads by ∼95%, memory usage by ∼90%, and improving database execution by 10–50×; designed a Node-based cron job to automate updates across the frontend and backend

Equity Research Analyst Jul 2025 – Present

Global Delta Securities *Remote*

- Conduct in-depth fundamental analysis of 2+ public equities, including financial statement modeling, industry benchmarking, and qualitative assessment
- Evaluate macroeconomic indicators, earnings reports, and sector trends to identify alpha-generating opportunities and downside risks
- Build and maintain DCF, comparable company, and precedent transaction valuation models to derive target prices and investment theses

Open-Source Developer Nov 2024 – Present

Self-Employed *Berkeley, CA*

- Resolved 10+ issues across 3+ open-source projects in programming systems, high-level analysis software, and instrumentation control (IoT) using React, TypeScript, and Python, some of which are used by NASA
- Provided constructive feedback on pull requests and documentation, creating 15+ test suites to enhance overall code quality and fostering community learning
- Collaborated with a global team of developers and issue trackers to triage bugs and discuss feature roadmaps

Projects

W3Check | *Next.js, TypeScript, Node.js, Tailwind CSS, PostgreSQL* Apr 2025 - Jul 2025

- Led full-stack development of W3Check, a web compliance platform automating WCAG verification and remediation workflows, winning $300 at a hackathon
- Engineered ∼90% of the codebase using Next.js, TypeScript, Node.js, and PostgreSQL, building scalable backend APIs and responsive frontend components
- Acted as technical point of contact for potential users, including UC Berkeley IT and the Sutardja Center for Entrepreneurship and Technology, translating user feedback into product improvements

Intro to Quant Finance Final Project | *Python, NumPy, Matplotlib, Pandas* Apr 2025

- Investigated different variations of swing trading utilizing simple moving averages
- Developed Python scripts to retrieve and analyze historical market data for backtesting purposes
- Integrated technical indicators (MACD) to optimize strategies, resulting in 8–17% backtested returns across selected timeframes and asset classes

Technical Skills

Languages: JavaScript, TypeScript, Python, Java, C, PostgreSQL
Frameworks: Next.js, React, Tailwind CSS, Bootstrap, Node.js, Flask, Django, Supabase, NumPy, Matplotlib, Pandas, TensorFlow, Scikit-learn
Developer Tools: Git, Docker
Certifications: Frontend Development, Machine Learning Foundations, Options 101, Generative AI